Wildfire Activity in Mediterranean Chile: Long-Term Burned-Area Dynamics, Ignition Causes, Initial Fuel, Burned Land Cover, and Municipal Fire Patterns

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Highlights

- Human-related ignition causes accounted for 99% of burned area with an identified cause.
- Two extreme fire seasons, 2016–2017 (566 thousand ha) and 2022–2023 (420 thousand ha), accounted for 43% of cumulative burned area during 2002–2003 to 2025–2026.
- Eleven municipalities recorded ≥2% of their area burned in at least six of the 24 fire seasons.
- Grasslands often dominated initial fuel. Plantations had the largest burned-area share.
- Regional and municipal patterns support geographically differentiated ignition prevention, fuel management, preparedness, and wildfire response.